

RINGKASAN HASIL P3 INFRASTRUKTUR JARINGAN ISP DAN PELANGGAN

Topologi: Internet Service Provider (ISP) & Routing Statis Environment: GNS3, Oracle VirtualBox, OS Host: Linux Mint

1. PENDAHULUAN & ARSITEKTUR JARINGAN

Praktikum ini bertujuan untuk membangun simulasi jaringan skala *Internet Service Provider* (ISP) dengan dua router pelanggan (Router A dan Router B) serta sebuah Ubuntu Server. Topologi ini memanfaatkan GNS3 sebagai *network simulator* dan VirtualBox sebagai *hypervisor* untuk menjalankan *Virtual Machines* (VM) MikroTik RouterOS dan Ubuntu.

Keseluruhan jaringan ini dirancang agar setiap klien (*Client A*, *Client B*, dan *Server*) dapat saling berkomunikasi (melalui implementasi *Static Routing*) dan terhubung ke internet global (melalui implementasi *NAT Masquerade*).

2. PENGATURAN INFRASTRUKTUR & ADAPTER JARINGAN (GNS3 & VIRTUALBOX)

Untuk memastikan komunikasi antar VM berjalan lancar di dalam ekosistem GNS3, pengaturan *Network Adapter* memerlukan perlakuan khusus.

A. Penggunaan Generic Driver (UDPTunnel)

- **Aturan Baku:** Seluruh sambungan kabel antar-Router (ISP ke Router Pelanggan) dan Router ke Client (Router ke Ubuntu) wajib menggunakan mode Generic Driver (UDPTunnel) pada VirtualBox.
- **Alasan teknis:** Mode *Internal Network* bawaan VirtualBox tidak akan berfungsi jika dihubungkan melalui kanvas GNS3. GNS3 bertindak sebagai perantara (makelar) paket data. GNS3 secara otomatis akan menyuntikkan port UDPTunnel (dengan parameter sport dan dport via 127.0.0.1) agar lalu lintas data dapat diatur sepenuhnya oleh GNS3. Oleh karena itu, penarikan kabel antar node cukup dilakukan langsung melalui GUI GNS3, dan VirtualBox akan menyesuaikan secara otomatis.

B. Setup Gateway Internet Menggunakan Cloud (virbr0) di Linux Mint

Untuk memberikan akses internet dari Wi-Fi *Host* (Linux Mint) ke dalam topologi GNS3, digunakan *node Cloud*.

- **Langkah Konfigurasi:**
 1. Tambahkan node Cloud ke dalam kanvas GNS3.
 2. Klik kanan pada Cloud → Configure.

3. Pada tab *Ethernet interfaces*, centang opsi "Show special Ethernet interfaces" (wajib dilakukan di Linux untuk menampilkan antarmuka virtual).
 4. Pilih antarmuka virbr0 dari menu *dropdown*, lalu klik Add dan OK.
 5. Hubungkan kabel dari Cloud (virbr0) ke Router ISP (Ethernet0).
- **Alasan Teknis:** virbr0 adalah antarmuka jembatan (*bridge virtual*) bawaan Linux yang sudah dilengkapi dengan fitur NAT dan DHCP secara *default*. Dengan menghubungkan Router ISP ke virbr0, Router ISP akan langsung mendapatkan suplai internet dari koneksi Wi-Fi *host* Linux Mint tanpa memicu pemblokiran *MAC Address* oleh *Access Point* fisik.

3. TABEL PENGALAMATAN IP (IP ADDRESSING)

Nama Perangkat	Interface GNS3	Interface Sistem	Alamat IP / Netmask	Keterangan / Gateway
Router ISP	Ethernet0	ether1	<i>DHCP Client</i>	Arah Cloud (virbr0) / Internet
	Ethernet1	ether2	200.200.200.1/24	Arah Ubuntu Server
	Ethernet2	ether3	100.100.100.1/30	Arah Router Pelanggan A
	Ethernet3	ether4	100.100.100.5/30	Arah Router Pelanggan B
Router A (R-PA)	Ethernet0	ether1	100.100.100.2/30	Arah Router ISP (GW: .1)
	Ethernet1	ether2	192.168.10.1/24	Arah Ubuntu Client A
Router B (R-PB)	Ethernet0	ether1	100.100.100.6/30	Arah Router ISP (GW: .5)
	Ethernet1	ether2	192.168.20.1/24	Arah Ubuntu Client B
Ubuntu Server	Ethernet0	enp0s3	200.200.200.2/24	GW: 200.200.200.1
Ubuntu Client A	Ethernet0	enp0s3	192.168.10.2/24	GW: 192.168.10.1
Ubuntu Client B	Ethernet0	enp0s3	192.168.20.2/24	GW: 192.168.20.1

4. KONFIGURASI ROUTER (MIKROTIK ROUTERS)

A. Konfigurasi Router ISP

1. Mendapatkan akses internet dari Cloud (virbr0)

/ip dhcp-client add interface=ether1 disabled=no

2. Konfigurasi IP Address Distribusi

/ip address add address=200.200.200.1/24 interface=ether2

/ip address add address=100.100.100.1/30 interface=ether3

/ip address add address=100.100.100.5/30 interface=ether4

3. Konfigurasi Routing Statis ke jaringan Pelanggan

/ip route add dst-address=192.168.10.0/24 gateway=100.100.100.2

/ip route add dst-address=192.168.20.0/24 gateway=100.100.100.6

4. Konfigurasi NAT agar seluruh jaringan di bawahnya dapat mengakses internet

/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade

B. Konfigurasi Router Pelanggan A (R-PA)

1. Konfigurasi IP Address

/ip address add address=100.100.100.2/30 interface=ether1

/ip address add address=192.168.10.1/24 interface=ether2

2. Konfigurasi Default Route ke arah ISP

/ip route add dst-address=0.0.0.0/0 gateway=100.100.100.1

3. Konfigurasi DNS

/ip dns set servers=8.8.8.8,8.8.4.4 allow-remote-requests=yes

4. Konfigurasi NAT untuk Client A

/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade

C. Konfigurasi Router Pelanggan B (R-PB)

1. Konfigurasi IP Address

/ip address add address=100.100.100.6/30 interface=ether1

/ip address add address=192.168.20.1/24 interface=ether2

2. Konfigurasi Default Route ke arah ISP

/ip route add dst-address=0.0.0.0/0 gateway=100.100.100.5

3. Konfigurasi DNS

/ip dns set servers=8.8.8.8,8.8.4.4 allow-remote-requests=yes

4. Konfigurasi NAT untuk Client B

/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade

5. KONFIGURASI CLIENT (UBUNTU NETPLAN)

A. Ubuntu Server

network:

version: 2

renderer: networkd

ethernets:

enp0s3:

addresses:

- 200.200.200.2/24

routes:

- to: default

via: 200.200.200.1

nameservers:

addresses: [8.8.8.8, 8.8.4.4]

B. Ubuntu Client A

network:

version: 2

renderer: networkd

ethernets:

enp0s3:

addresses:

- 192.168.10.2/24

routes:

- to: default

via: 192.168.10.1

nameservers:

addresses: [8.8.8.8, 8.8.4.4]

C. Ubuntu Client B

network:

version: 2

renderer: networkd

ethernets:

enp0s3:

addresses:

- 192.168.20.2/24

routes:

- to: default

via: 192.168.20.1

nameservers:

addresses: [8.8.8.8, 8.8.4.4]

6. HASIL PENGUJIAN & VERIFIKASI

Tahapan pengujian menunjukkan hasil yang sukses:

1. **Uji Konektivitas Gateway:** Seluruh Ubuntu *Client* berhasil melakukan *ping* ke alamat IP masing-masing antarmuka router terdekat.
2. **Uji Routing Antar-Jaringan:** Klien A (192.168.10.2) berhasil melakukan *ping* ke Klien B (192.168.20.2) dan Ubuntu Server (200.200.200.2), membuktikan bahwa tabel *routing* pada Router ISP bekerja dengan baik.
3. **Uji Akses Internet Global:** Seluruh *router* dan Ubuntu *client* berhasil melakukan *ping* ke alamat DNS eksternal (ping 8.8.8.8 dan ping google.com), memastikan fungsionalitas NAT dan fitur *bridging* dari *virbr0* berfungsi optimal.
4. **Uji Jalur Paket:** Seluruh VM Ubuntu mulai dari Client A dan B hingga Server harus bisa menunjuk berapa jalur paket router yang dilewati (hop) untuk mendapatkan internet dengan sempurna

Tahap 1: Tes Koneksi ke Gateway Terdekat (Lokal)

1.1 Di Terminal Ubuntu Client A:

ping 192.168.10.1

```
client@client:~$ ping 192.168.10.1
PING 192.168.10.1 (192.168.10.1) 56(84) bytes of data.
64 bytes from 192.168.10.1: icmp_seq=1 ttl=64 time=0.937 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=64 time=0.720 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=64 time=1.74 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=64 time=1.06 ms
64 bytes from 192.168.10.1: icmp_seq=5 ttl=64 time=2.05 ms
64 bytes from 192.168.10.1: icmp_seq=6 ttl=64 time=0.569 ms
^C
--- 192.168.10.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5130ms
rtt min/avg/max/mdev = 0.569/1.177/2.046/0.535 ms
```

1.2 Di Terminal Ubuntu Client B:

ping 192.168.20.1

```
client@client:~$ ping 192.168.20.1
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=64 time=0.551 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=64 time=0.558 ms
^C
--- 192.168.20.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1201ms
rtt min/avg/max/mdev = 0.551/0.554/0.558/0.003 ms
```

1.3 Di Terminal Ubuntu Server:

ping 200.200.200.1

```
server@server:~$ ping 200.200.200.1
PING 200.200.200.1 (200.200.200.1) 56(84) bytes of data.
64 bytes from 200.200.200.1: icmp_seq=1 ttl=64 time=0.964 ms
64 bytes from 200.200.200.1: icmp_seq=2 ttl=64 time=0.611 ms
64 bytes from 200.200.200.1: icmp_seq=3 ttl=64 time=0.897 ms
64 bytes from 200.200.200.1: icmp_seq=4 ttl=64 time=0.605 ms
^C
--- 200.200.200.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3049ms
rtt min/avg/max/mdev = 0.605/0.769/0.964/0.162 ms
```

Tahap 2: Tes Koneksi Antar Pelanggan (Uji Routing Statis ISP)

2.1 Di Terminal Ubuntu Client A, coba *ping* ke IP Client B:

ping 192.168.20.2

```
client@client:~$ ping 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=61 time=1.93 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=61 time=2.61 ms

^C--- 192.168.20.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1214ms
rtt min/avg/max/mdev = 1.929/2.270/2.612/0.341 ms
```

2.2 Di Terminal Ubuntu Client B, coba *ping* ke IP Server:

ping 200.200.200.2

```
client@client:~$ ping 200.200.200.2
PING 200.200.200.2 (200.200.200.2) 56(84) bytes of data.
64 bytes from 200.200.200.2: icmp_seq=1 ttl=62 time=1.89 ms
64 bytes from 200.200.200.2: icmp_seq=2 ttl=62 time=1.22 ms
64 bytes from 200.200.200.2: icmp_seq=3 ttl=62 time=1.21 ms
64 bytes from 200.200.200.2: icmp_seq=4 ttl=62 time=1.55 ms
64 bytes from 200.200.200.2: icmp_seq=5 ttl=62 time=1.71 ms
64 bytes from 200.200.200.2: icmp_seq=6 ttl=62 time=1.43 ms
64 bytes from 200.200.200.2: icmp_seq=7 ttl=62 time=2.35 ms
64 bytes from 200.200.200.2: icmp_seq=8 ttl=62 time=1.37 ms

^C
--- 200.200.200.2 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7050ms
rtt min/avg/max/mdev = 1.206/1.588/2.346/0.359 ms
```


Tahap 3: Tes Koneksi Jagat Raya (Internet)

3.1 Di semua Ubuntu (Client A, Client B, dan Server):

ping google.com

```
client@client:~$ ping google.com
PING google.com (172.253.118.113) 56(84) bytes of data.
64 bytes from sl-in-f113.1e100.net (172.253.118.113): icmp_seq=1 ttl=58 time=34.0 ms
64 bytes from sl-in-f113.1e100.net (172.253.118.113): icmp_seq=2 ttl=58 time=26.0 ms
64 bytes from sl-in-f113.1e100.net (172.253.118.113): icmp_seq=3 ttl=58 time=18.1 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2091ms
rtt min/avg/max/mdev = 18.146/26.038/34.006/6.475 ms
```

```
client@client:~$ ping -c 5 google.com
PING google.com (172.253.118.102) 56(84) bytes of data.
64 bytes from sl-in-f102.1e100.net (172.253.118.102): icmp_seq=1 ttl=58 time=11.7 ms
64 bytes from sl-in-f102.1e100.net (172.253.118.102): icmp_seq=2 ttl=58 time=17.8 ms
64 bytes from sl-in-f102.1e100.net (172.253.118.102): icmp_seq=3 ttl=58 time=14.0 ms
64 bytes from sl-in-f102.1e100.net (172.253.118.102): icmp_seq=4 ttl=58 time=12.4 ms
64 bytes from sl-in-f102.1e100.net (172.253.118.102): icmp_seq=5 ttl=58 time=18.9 ms
--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4143ms
rtt min/avg/max/mdev = 11.700/14.955/18.865/2.870 ms
```

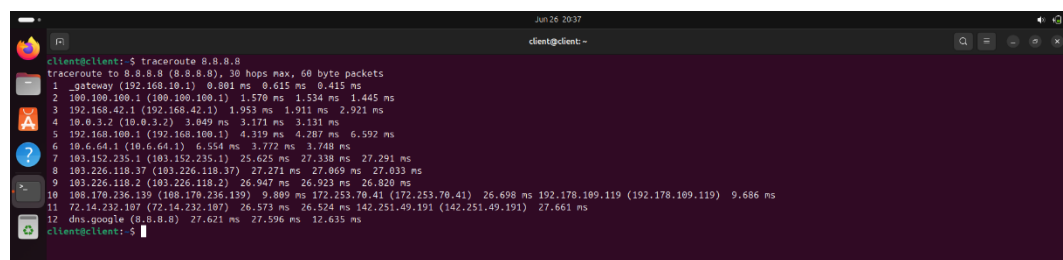
```
[admin@MikroTik] > /ping google.com
SEQ HOST                               SIZE TTL TIME                               STATUS
0 172.253.118.113                       56 253 209ms994us
sent=1 received=1 packet-loss=0% min-rtt=209ms994us avg-rtt=209ms994us
max-rtt=209ms994us
```

Tahap 4: Tes Jalur Paket (Traceroute) di VM Ubuntu Client A dan B serta Server:

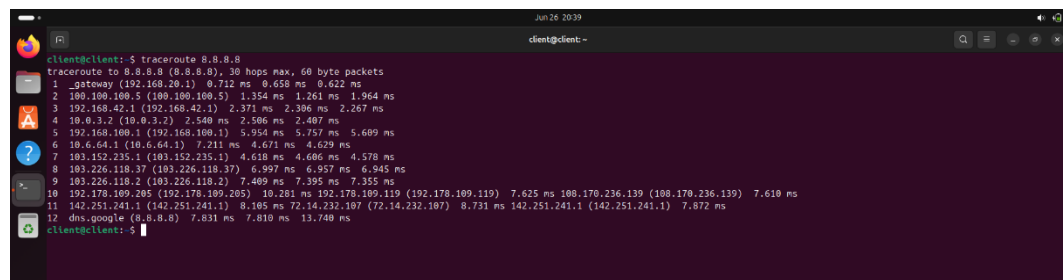
4.1 Di semua Ubuntu (Client A, Client B, dan Server):

traceroute 8.8.8.8

```
server@server:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1 _gateway (200.200.200.1) 0.784 ms 0.724 ms 0.702 ms
 2 192.168.42.1 (192.168.42.1) 1.779 ms 1.757 ms 1.742 ms
 3 10.0.3.2 (10.0.3.2) 1.980 ms 1.966 ms 1.951 ms
 4 192.168.1.1 (192.168.1.1) 3.944 ms 3.930 ms 3.916 ms
 5 10.9.30.254 (10.9.30.254) 10.220 ms 10.205 ms 10.192 ms
 6 FAST-INTERNET-202-154-184-253.solnet.net.id (202.154.184.253) 9.781 ms 3.622 ms 3.588 ms
 7 FAST-INTERNET-223-130-21-181.solnet.net.id (223.130.21.181) 6.341 ms 6.067 ms 6.019 ms
 8 FAST-INTERNET-223-130-21-121.solnet.net.id (223.130.21.121) 6.448 ms 6.407 ms 6.392 ms
 9 192.178.71.194 (192.178.71.194) 6.561 ms 6.566 ms 6.693 ms
10 * * *
11 dns.google (8.8.8.8) 8.131 ms 8.087 ms 6.854 ms
server@server:~$
```



```
Jun 26 20:37
client@client: ~
client@client:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1 _gateway (192.168.10.1) 0.501 ms 0.615 ms 0.415 ms
 2 100.100.100.1 (100.100.100.1) 1.570 ms 1.534 ms 1.445 ms
 3 192.168.42.1 (192.168.42.1) 1.953 ms 1.911 ms 2.921 ms
 4 10.0.3.2 (10.0.3.2) 3.049 ms 3.171 ms 3.131 ms
 5 192.168.100.1 (192.168.100.1) 4.319 ms 4.287 ms 6.592 ms
 6 10.6.64.1 (10.6.64.1) 6.554 ms 3.772 ms 3.748 ms
 7 103.152.235.1 (103.152.235.1) 25.625 ms 27.338 ms 27.291 ms
 8 103.226.118.37 (103.226.118.37) 27.271 ms 27.069 ms 27.033 ms
 9 103.226.118.2 (103.226.118.2) 26.947 ms 26.923 ms 26.920 ms
10 100.170.236.139 (100.170.236.139) 9.889 ms 172.253.70.41 (172.253.70.41) 26.698 ms 192.178.109.119 (192.178.109.119) 9.686 ms
11 72.14.232.107 (72.14.232.107) 26.573 ms 26.524 ms 142.251.49.191 (142.251.49.191) 27.661 ms
12 dns.google (8.8.8.8) 27.621 ms 27.596 ms 12.635 ms
client@client:~$
```



```
Jun 26 20:39
client@client: ~
client@client:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1 _gateway (192.168.20.1) 0.712 ms 0.658 ms 0.622 ms
 2 100.100.100.5 (100.100.100.5) 1.354 ms 1.261 ms 1.964 ms
 3 192.168.42.1 (192.168.42.1) 2.371 ms 2.306 ms 2.267 ms
 4 10.0.3.2 (10.0.3.2) 2.540 ms 2.506 ms 2.407 ms
 5 192.168.100.1 (192.168.100.1) 5.954 ms 5.757 ms 5.609 ms
 6 10.6.64.1 (10.6.64.1) 7.211 ms 4.671 ms 4.629 ms
 7 103.152.235.1 (103.152.235.1) 4.618 ms 4.666 ms 4.578 ms
 8 103.226.118.37 (103.226.118.37) 6.997 ms 6.957 ms 6.945 ms
 9 103.226.118.2 (103.226.118.2) 7.409 ms 7.395 ms 7.355 ms
10 192.178.109.205 (192.178.109.205) 10.281 ms 192.178.109.119 (192.178.109.119) 7.625 ms 100.170.236.139 (100.170.236.139) 7.610 ms
11 142.251.241.1 (142.251.241.1) 8.105 ms 72.14.232.107 (72.14.232.107) 8.731 ms 142.251.241.1 (142.251.241.1) 7.872 ms
12 dns.google (8.8.8.8) 7.831 ms 7.810 ms 13.740 ms
client@client:~$
```